

M004 · A Space that Reacts — Event → Check → Action — English Worksheet

Topic: A Space that Reacts — Event → Check → Action · **Course:** Functions & Games · **Time:** about 45 minutes

This lesson your world **reacts to the player**. The pattern: an **event** happens → the code **checks** something → an **action** runs.

1 · Predict

```
on player walks on gold block:
    say "Welcome to my world!"
```

When does the message appear? What happens when the player walks on grass instead?


```
on player walks on stone bricks:
    if block ahead is iron door:
        open the door
    otherwise:
        say "No door here."
```

The player walks on stone bricks right in front of the iron door. What happens? What happens on stone bricks far away from the door?


```
function spring_trap():  
    replace block below with air  
    say "You fell in my trap!"  
  
on player walks on red carpet:  
    spring_trap()
```

What happens when the player steps on red carpet? Why is the trap written as a function?

2 · Spot the Bug 🐛

Each block below is broken. Rewrite it so it works, then explain why the original was wrong and why your fix works.

Bug A — The door should open **only after** the code checks that there really is an iron door ahead. Right now the check and the action are **swapped**.

```
on player walks on stone bricks:  
    open the door  
    if block ahead is iron door:  
        say "Door found!"
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The trap is supposed to fire when the player steps on **red carpet**. But in the world, the red carpet does nothing and the blue carpet — the safe path — keeps eating players.

```
on player walks on blue carpet:  
    spring_trap()
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The secret message should appear **only** when the player stands on a gold block. Right now the message appears everywhere, on every single step.

```
on player walks:  
  say "You found the secret room!"
```

Hint: the event fires on every step — the **check** is missing.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Switch to your homework world. Build a space that reacts to the player — use the event → check → action pattern.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.